Homework 6

Problem 1. Determine x_1 and x_2 so as to

$$\begin{aligned} \max \quad & Z = 12x_1 + 21x_2 + 2x_1x_2 - 2x_1^2 - 2x_2^2 \\ \text{s.t.} \quad & x_2 \le 8, \\ & x_1 + x_2 \le 10, \\ & x_1, x_2 \ge 0. \end{aligned}$$

Solution. [**Standard**]

We first rewrite the maximization problem as the following minimization problem:

$$\min \quad \phi(x_1, x_2) = -12x_1 - 21x_2 - 2x_1x_2 + 2x_1^2 + 2x_2^2.$$

The constraints can be written as

$$8 - x_2 \ge 0, \qquad 10 - x_1 - x_2 \ge 0, \qquad x_1 \ge 0, \qquad x_2 \ge 0.$$

The Lagrange function is

$$\begin{aligned} L = & -12x_1 - 21x_2 - 2x_1x_2 + 2x_1^2 + 2x_2^2 \\ & - \mu_1(8 - x_2) - \mu_2(10 - x_1 - x_2) - s_1x_1 - s_2x_2. \end{aligned}$$

Here

$$\mu_1, \mu_2, s_1, s_2 \ge 0.$$

The KKT conditions consist of the following four types of conditions.

1. Lagrange derivative condition (LDC):

$$\frac{\partial L}{\partial x_1} = 0 \quad \Longrightarrow \quad -12 - 2x_2 + 4x_1 + \mu_2 - s_1 = 0,$$

$$\frac{\partial L}{\partial x_2} = 0 \quad \Longrightarrow \quad -21 - 2x_1 + 4x_2 + \mu_1 + \mu_2 - s_2 = 0.$$

2. Primal feasibility condition, or original decision constraints (ODC):

$$8 - x_2 \ge 0,$$

$$10 - x_1 - x_2 \geq 0,$$
$$x_1 \geq 0, \qquad x_2 \geq 0.$$

3. Dual feasibility condition, or multiplier sign conditions (MSC):

$$\mu_1 \geq 0, \qquad \mu_2 \geq 0, \qquad s_1 \geq 0, \qquad s_2 \geq 0.$$

4. Complementary slackness condition (CSC):

$$\mu_1(8 - x_2) = 0,$$
$$\mu_2(10 - x_1 - x_2) = 0,$$
$$s_1 x_1 = 0,$$
$$s_2 x_2 = 0.$$

We now solve the KKT system.

First, we show that $x_1 > 0$ and $x_2 > 0$.

Suppose $x_1 = 0$. Then the first stationarity condition gives

$$-12 - 2x_2 + \mu_2 - s_1 = 0,$$

or equivalently

$$s_1 = \mu_2 - 12 - 2x_2.$$

Since $s_1 \geq 0$, we must have

$$\mu_2 \geq 12 + 2x_2 > 0.$$

Thus $\mu_2 > 0$. By complementary slackness,

$$10 - x_1 - x_2 = 0.$$

Since $x_1 = 0$, this implies

$$x_2 = 10.$$

However, this violates the constraint $x_2 \leq 8$. Hence $x_1 = 0$ is impossible, and therefore

$$x_1 > 0.$$

Suppose $x_2 = 0$. Since $8 - x_2 = 8 > 0$, complementary slackness gives

$$\mu_1 = 0.$$

From the previous paragraph, $x_1 > 0$, so complementary slackness gives

$$s_1 = 0.$$

The first stationarity condition becomes

$$-12 + 4x_1 + \mu_2 = 0.$$

If $\mu_2 > 0$, then by complementary slackness,

$$10 - x_1 - x_2 = 0.$$

Since $x_2 = 0$, this gives $x_1 = 10$. Substituting into

$$-12 + 4x_1 + \mu_2 = 0$$

gives

$$\mu_2 = -28,$$

which contradicts $\mu_2 \geq 0$. Hence $\mu_2 = 0$.

Therefore,

$$-12 + 4x_1 = 0,$$

so

$$x_1 = 3.$$

Now the second stationarity condition becomes

$$-21 - 2(3) - s_2 = 0,$$

so

$$s_2 = -27,$$

which contradicts $s_2 \geq 0$. Hence $x_2 = 0$ is impossible, and therefore

$$x_2 > 0.$$

Since

$$x_1 > 0, \qquad x_2 > 0,$$

the complementary slackness conditions imply

$$s_1 = s_2 = 0.$$

Thus the stationarity conditions reduce to

$$-12 - 2x_2 + 4x_1 + \mu_2 = 0,$$

$$-21 - 2x_1 + 4x_2 + \mu_1 + \mu_2 = 0.$$

We now consider the possible active sets of the two constraints

$$8 - x_2 \geq 0, \qquad 10 - x_1 - x_2 \geq 0.$$

Case 1: $\mu_1 = 0, \mu_2 = 0$.

Then

$$-12 - 2x_2 + 4x_1 = 0,$$
$$-21 - 2x_1 + 4x_2 = 0.$$

Solving gives

$$x_1 = \frac{15}{2}, \qquad x_2 = 9.$$

This violates the constraint $x_2 \leq 8$. Hence this case is infeasible.

Case 2: $\mu_1 > 0, \mu_2 > 0$.

By complementary slackness,

$$8 - x_2 = 0, \qquad 10 - x_1 - x_2 = 0.$$

Thus

$$x_2 = 8, \qquad x_1 = 2.$$

Substituting into the first stationarity condition gives

$$-12 - 2(8) + 4(2) + \mu_2 = 0,$$

so

$$\mu_2 = 20.$$

Substituting into the second stationarity condition gives

$$-21 - 2(2) + 4(8) + \mu_1 + 20 = 0.$$

Hence

$$\mu_1 = -27.$$

This violates $\mu_1 \geq 0$. Hence this case is infeasible.

Case 3: $\mu_1 > 0, \mu_2 = 0$.

By complementary slackness,

$$8 - x_2 = 0,$$

so

$$x_2 = 8.$$

The first stationarity condition gives

$$-12 - 2(8) + 4x_1 = 0,$$

so

$$x_1 = 7.$$

But then

$$x_1 + x_2 = 15 > 10,$$

which violates the constraint $x_1 + x_2 \leq 10$. Hence this case is infeasible.

Case 4: $\mu_1 = 0, \mu_2 > 0$.

By complementary slackness,

$$10 - x_1 - x_2 = 0,$$

so

$$x_1 + x_2 = 10.$$

The stationarity conditions become

$$-12 - 2x_2 + 4x_1 + \mu_2 = 0,$$

$$-21 - 2x_1 + 4x_2 + \mu_2 = 0.$$

From the first equation,

$$\mu_2 = 12 + 2x_2 - 4x_1.$$

From the second equation,

$$\mu_2 = 21 + 2x_1 - 4x_2.$$

Therefore,

$$12 + 2x_2 - 4x_1 = 21 + 2x_1 - 4x_2.$$

Thus

$$6x_2 - 6x_1 = 9,$$

or

$$x_2 - x_1 = \frac{3}{2}.$$

Combining this with

$$x_1 + x_2 = 10,$$

we get

$$x_1 = \frac{17}{4}, \qquad x_2 = \frac{23}{4}.$$

Then

$$\mu_2 = 12 + 2 \cdot \frac{23}{4} - 4 \cdot \frac{17}{4} = \frac{13}{2}.$$

Hence

$$\mu_1 = 0, \qquad \mu_2 = \frac{13}{2}.$$

Since

$$\mu_1 \geq 0, \qquad \mu_2 \geq 0,$$

this case satisfies all KKT conditions.

Therefore,

$$x_1^* = \frac{17}{4}, \qquad x_2^* = \frac{23}{4}.$$

Finally, we verify that this KKT point is the global maximizer. The Hessian matrix of the original objective function

$$Z = 12x_1 + 21x_2 + 2x_1x_2 - 2x_1^2 - 2x_2^2$$

is

$$A = \nabla^2 Z = \begin{pmatrix} -4 & 2 \\ 2 & -4 \end{pmatrix}.$$

Then

$$-A = \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}.$$

The leading principal minors of $-A$ are

$$\Delta_1 = 4 > 0,$$

and

$$\Delta_2 = \begin{vmatrix} 4 & -2 \\ -2 & 4 \end{vmatrix} = 16 - 4 = 12 > 0.$$

Therefore, $-A$ is positive definite. Hence $A = \nabla^2 Z$ is negative definite.

Thus Z is strictly concave. Since constraints are linear, the feasible region is convex, this KKT point is the global maximizer. Therefore,

$$x_1^* = \frac{17}{4}, \qquad x_2^* = \frac{23}{4}.$$

The optimal value is

$$\begin{aligned} Z^* &= 12 \cdot \frac{17}{4} + 21 \cdot \frac{23}{4} + 2 \cdot \frac{17}{4} \cdot \frac{23}{4} - 2\left(\frac{17}{4}\right)^2 - 2\left(\frac{23}{4}\right)^2 \\ &= \frac{947}{8}. \end{aligned}$$

Hence,

$$x^* = \left(\frac{17}{4}, \frac{23}{4}\right), \qquad Z^* = \frac{947}{8}.$$

□

Solution. [**Shortcut**]

We first rewrite the maximization problem as the following minimization problem:

$$\min \quad \phi(x_1, x_2) = -12x_1 - 21x_2 - 2x_1x_2 + 2x_1^2 + 2x_2^2.$$

The constraints can be written as

$$8 - x_2 \geq 0, \qquad 10 - x_1 - x_2 \geq 0, \qquad x_1 \geq 0, \qquad x_2 \geq 0.$$

We use the shortcut form of the KKT conditions for the nonnegative variables x_1, x_2. Thus the nonnegativity constraints are not written explicitly in the Lagrange function.

The Lagrange function is

$$\begin{aligned} L = &- 12x_1 - 21x_2 - 2x_1x_2 + 2x_1^2 + 2x_2^2 \\ &- \mu_1(8 - x_2) - \mu_2(10 - x_1 - x_2). \end{aligned}$$

Here

$$\mu_1, \mu_2 \geq 0.$$

The KKT conditions consist of the following four types of conditions.

1. Lagrange derivative condition (LDC):

For $x_1 \geq 0$, we have

$$\frac{\partial L}{\partial x_1} \geq 0, \qquad x_1 \frac{\partial L}{\partial x_1} = 0.$$

Since

$$\frac{\partial L}{\partial x_1} = -12 - 2x_2 + 4x_1 + \mu_2,$$

we obtain

$$-12 - 2x_2 + 4x_1 + \mu_2 \geq 0,$$
$$x_1(-12 - 2x_2 + 4x_1 + \mu_2) = 0.$$

For $x_2 \geq 0$, we have

$$\frac{\partial L}{\partial x_2} \geq 0, \qquad x_2 \frac{\partial L}{\partial x_2} = 0.$$

Since

$$\frac{\partial L}{\partial x_2} = -21 - 2x_1 + 4x_2 + \mu_1 + \mu_2,$$

we obtain

$$-21 - 2x_1 + 4x_2 + \mu_1 + \mu_2 \geq 0,$$
$$x_2(-21 - 2x_1 + 4x_2 + \mu_1 + \mu_2) = 0.$$

2. Primal feasibility condition, or original decision constraints (ODC):

$$8 - x_2 \geq 0,$$

$$10 - x_1 - x_2 \geq 0,$$

$$x_1 \geq 0, \qquad x_2 \geq 0.$$

3. Dual feasibility condition, or multiplier sign conditions (MSC):

$$\mu_1 \geq 0, \qquad \mu_2 \geq 0.$$

4. Complementary slackness condition (CSC):

$$\mu_1(8 - x_2) = 0,$$

$$\mu_2(10 - x_1 - x_2) = 0.$$

We now solve the KKT system.

First, we show that $x_1 > 0$ and $x_2 > 0$.

Suppose $x_1 = 0$. Then, by primal feasibility,

$$x_2 \leq 8.$$

Hence

$$10 - x_1 - x_2 = 10 - x_2 \geq 2 > 0.$$

By complementary slackness,

$$\mu_2 = 0.$$

The first Lagrange derivative condition becomes

$$-12 - 2x_2 \geq 0,$$

which is impossible since $x_2 \geq 0$. Therefore,

$$x_1 > 0.$$

Suppose $x_2 = 0$. Since

$$8 - x_2 = 8 > 0,$$

complementary slackness gives

$$\mu_1 = 0.$$

If $x_1 < 10$, then

$$10 - x_1 - x_2 = 10 - x_1 > 0,$$

so

$$\mu_2 = 0.$$

The second Lagrange derivative condition becomes

$$-21 - 2x_1 \geq 0,$$

which is impossible since $x_1 \geq 0$.

If $x_1 = 10$, then $x_1 > 0$. Hence

$$x_1 \frac{\partial L}{\partial x_1} = 0$$

implies

$$\frac{\partial L}{\partial x_1} = 0.$$

That is,

$$-12 - 2(0) + 4(10) + \mu_2 = 0,$$

so

$$28 + \mu_2 = 0,$$

which contradicts $\mu_2 \geq 0$.

Therefore, $x_2 = 0$ is impossible, and hence

$$x_2 > 0.$$

Since

$$x_1 > 0, \qquad x_2 > 0,$$

the shortcut complementary conditions imply

$$\frac{\partial L}{\partial x_1} = 0, \qquad \frac{\partial L}{\partial x_2} = 0.$$

Therefore,

$$-12 - 2x_2 + 4x_1 + \mu_2 = 0,$$

$$-21 - 2x_1 + 4x_2 + \mu_1 + \mu_2 = 0.$$

Now we consider the possible active sets of the two constraints

$$8 - x_2 \geq 0, \qquad 10 - x_1 - x_2 \geq 0.$$

Case 1: $\mu_1 = 0, \mu_2 = 0$.

Then

$$-12 - 2x_2 + 4x_1 = 0,$$

$$-21 - 2x_1 + 4x_2 = 0.$$

Solving gives

$$x_1 = \frac{15}{2}, \qquad x_2 = 9.$$

This violates the constraint $x_2 \leq 8$. Hence this case is infeasible.

Case 2: $\mu_1 > 0, \mu_2 > 0$.

By complementary slackness,

$$8 - x_2 = 0, \qquad 10 - x_1 - x_2 = 0.$$

Thus

$$x_2 = 8, \qquad x_1 = 2.$$

Substituting into the first equation gives

$$-12 - 2(8) + 4(2) + \mu_2 = 0,$$

so

$$\mu_2 = 20.$$

Substituting into the second equation gives

$$-21 - 2(2) + 4(8) + \mu_1 + 20 = 0.$$

Hence

$$\mu_1 = -27.$$

This violates $\mu_1 \geq 0$. Hence this case is infeasible.

Case 3: $\mu_1 > 0, \mu_2 = 0$.

By complementary slackness,

$$8 - x_2 = 0,$$

so

$$x_2 = 8.$$

The first equation gives

$$-12 - 2(8) + 4x_1 = 0,$$

so

$$x_1 = 7.$$

But then

$$x_1 + x_2 = 15 > 10,$$

which violates the constraint $x_1 + x_2 \leq 10$. Hence this case is infeasible.

Case 4: $\mu_1 = 0, \mu_2 > 0$.

By complementary slackness,

$$10 - x_1 - x_2 = 0,$$

so

$$x_1 + x_2 = 10.$$

The two equations become

$$-12 - 2x_2 + 4x_1 + \mu_2 = 0,$$

$$-21 - 2x_1 + 4x_2 + \mu_2 = 0.$$

From the first equation,

$$\mu_2 = 12 + 2x_2 - 4x_1.$$

From the second equation,

$$\mu_2 = 21 + 2x_1 - 4x_2.$$

Therefore,

$$12 + 2x_2 - 4x_1 = 21 + 2x_1 - 4x_2.$$

Thus

$$6x_2 - 6x_1 = 9,$$

or

$$x_2 - x_1 = \frac{3}{2}.$$

Combining this with

$$x_1 + x_2 = 10,$$

we obtain

$$x_1 = \frac{17}{4}, \qquad x_2 = \frac{23}{4}.$$

Then

$$\mu_2 = 12 + 2 \cdot \frac{23}{4} - 4 \cdot \frac{17}{4} = \frac{13}{2}.$$

Hence

$$\mu_1 = 0, \qquad \mu_2 = \frac{13}{2}.$$

This satisfies all KKT conditions.

Therefore,

$$x_1^* = \frac{17}{4}, \qquad x_2^* = \frac{23}{4}.$$

Finally, we verify that this KKT point is the global maximizer. The Hessian matrix of the original objective function

$$Z = 12x_1 + 21x_2 + 2x_1x_2 - 2x_1^2 - 2x_2^2$$

is

$$A = \nabla^2 Z = \begin{pmatrix} -4 & 2 \\ 2 & -4 \end{pmatrix}.$$

Then

$$-A = \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}.$$

The leading principal minors of $-A$ are

$$\Delta_1 = 4 > 0,$$

and

$$\Delta_2 = \begin{vmatrix} 4 & -2 \\ -2 & 4 \end{vmatrix} = 16 - 4 = 12 > 0.$$

Therefore, $-A$ is positive definite. Hence $A = \nabla^2 Z$ is negative definite.

Thus Z is strictly concave. Since constraints are linear, the feasible region is convex, this KKT point is the global maximizer.

The optimal value is

$$\begin{aligned} Z^* &= 12 \cdot \frac{17}{4} + 21 \cdot \frac{23}{4} + 2 \cdot \frac{17}{4} \cdot \frac{23}{4} - 2\left(\frac{17}{4}\right)^2 - 2\left(\frac{23}{4}\right)^2 \\ &= \frac{947}{8}. \end{aligned}$$

Hence,

$$x^* = \left(\frac{17}{4}, \frac{23}{4}\right), \qquad Z^* = \frac{947}{8}.$$

□